

Pre-Registered Experimental Protocol: Adaptive Memory Ablations

Testing whether Stage One outcomes are mechanism-specific or parameter-specific

Core Research Document - Next confirmatory experiment

Adaptive Memory Dynamics Research Program Stage One revision - September 2026

Evidence status: Stage One is a reproducible sklearn-digits-8x8 experiment across five independent seeds. Reinforced retention approximately matches the persistent control on ordinary accuracy and Task B adaptation; passive decay reduces accuracy; neither decay mechanism improves Task A retention in the current sequential setup.

1. Rationale

Stage One produced mixed/negative evidence. The next experiment therefore tests decay rate, reinforcement strength, and task-sequence sensitivity before any broader claim about adaptive forgetting is made.

2. Confirmatory hypotheses

ID	Hypothesis
H0-1	No tested adaptive-memory setting improves the pre-specified retention-adaptation criterion over persistent retention.
H1-1	At least one pre-specified adaptive-memory setting reproducibly improves that criterion.
H0-2	Observed Stage One mechanism ordering is stable across task sequences.
H1-2	Mechanism effects interact materially with task sequence or inactivity duration.

3. Fixed controls

- Identical model capacity, optimizer family, batch size, training budget, preprocessing, and data splits within a comparison.
- Independent training seeds recorded and preserved.
- Persistent baseline included in every benchmark/task-sequence block.
- Hyperparameter-discovery runs separated from final confirmatory evaluation.

4. Primary outcomes

Primary: prior-task forgetting and new-task accuracy. Secondary: final held-out accuracy, post-idle accuracy, adaptation speed, weight norm, mean retention, and representation drift when instrumented.

5. Statistical discipline

- Use training seed as the independent replication unit.
- Report means, sample SD/intervals, and effect sizes.
- Do not remove poor-performing seeds absent a documented software/data failure.
- Control multiple comparisons across rate/strength sweeps.
- Freeze analysis choices before interpreting the confirmatory benchmark.

Figure 1. Frozen Stage One reference accuracy. These values motivate, but are not part of, the future confirmatory ablation.

Figure 2. Frozen Stage One Task A forgetting reference.

6. Decision rule

The study will claim a useful adaptive-memory regime only if a pre-specified improvement is reproducible and does not rely on post-hoc selection from the same evaluation seeds. Otherwise the result will be reported as neutral, harmful, or task-dependent.

Figures

